Adaptive Experimental Design: Pointwise Information, the Ignorance Gap, and Joint Geometry–Policy Optimization

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April 2026

Abstract

We develop a self-contained theoretical framework for the joint optimization of sensor hardware $\boldsymbol{c}$ and adaptive measurement policy π under a Bayesian prior on the unknown state $\boldsymbol{x}$. The framework is organized around the *pointwise expected information gain* $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$—the expected reduction in posterior entropy about $\boldsymbol{x}$ under observations drawn from the true-$\boldsymbol{x}$ likelihood. Its prior average is the standard mutual information $I(\boldsymbol{x}; \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})$, and its pointwise evaluation cleanly defines the clairvoyant *oracle* $V_{\text{oracle}}(\boldsymbol{x}) = \max_{\boldsymbol{s}} \Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, the pre-committed *fixed design* $V_{\text{fixed}}(\boldsymbol{c}) = \max_{\boldsymbol{s}} \mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})]$, the *ignorance gap* $IG(\boldsymbol{x}) = V_{\text{oracle}}(\boldsymbol{x}) - \Phi(\boldsymbol{x}, \boldsymbol{s}^{\star}_{\text{fixed}}, \boldsymbol{c})$, and the adaptive value $V_{\text{adaptive}}(\boldsymbol{c})$ via Bellman dynamic programming on the belief state. The prior-expected ignorance gap $\mathbb{E}[IG]$ upper-bounds the adaptive advantage $V_{\text{adaptive}} - V_{\text{fixed}}$—the classical Expected Value of Perfect Information (EVPI). We show this bound is typically loose and, crucially, that $\operatorname{argmax}_{\boldsymbol{c}} \mathbb{E}[IG](\boldsymbol{c}) \neq \operatorname{argmax}_{\boldsymbol{c}} V_{\text{adaptive}}(\boldsymbol{c})$: EVPI is a go/no-go diagnostic, not a design target. We formulate the joint $(\boldsymbol{c}, \pi)$ optimization, derive the envelope-theorem geometry gradient that avoids differentiating through the policy's argmax, and prescribe a policy-iteration scheme that alternates Bellman policy solves with gradient steps on $\boldsymbol{c}$. A fully-solved radar beam-search case study exhibits every ingredient at minimum dimension, including a quantitative $\sim 2.8\times$ separation between EVPI-guided and joint-DP-guided geometry selection. A final section catalogs principled relaxations when the exact Bellman DP is intractable, ordered by structure preserved. An appendix derives the Bellman equation from first principles.

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1 Introduction

A sensor designed in isolation and a measurement policy designed in isolation are both half-designed. When the operator of the sensor will react to intermediate observations—steering an antenna, changing a pulse delay, reconfiguring a metasurface—the optimal hardware depends on what the operator will actually do, and the optimal operator strategy depends on what the hardware allows. This paper develops a clean formulation of that joint problem.

The organizing quantity throughout is not Fisher information, not the Cramér–Rao bound, not the posterior variance, but a single information-theoretic functional: the *pointwise expected information gain* $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, defined as the expected reduction in posterior entropy about the unknown state $\boldsymbol{x}$ when the measurement is performed with sensor parameters $\boldsymbol{s}$ and device geometry $\boldsymbol{c}$, *given* that the true state is $\boldsymbol{x}$. Its prior average is the standard mutual information between $\boldsymbol{x}$ and the data. Its pointwise evaluation—holding $\boldsymbol{x}$ fixed—lets us define a clairvoyant oracle that knows $\boldsymbol{x}$, a non-adaptive fixed design that must hedge across the prior, and an adaptive policy that reacts to intermediate data. The hierarchy

$$V_{\text{fixed}}(\boldsymbol{c}) \;\le\; V_{\text{adaptive}}(\boldsymbol{c}) \;\le\; \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x};\, \boldsymbol{c})]$$

is exact. The gap between the first two is the quantity adaptation can win; the gap between the first and third is the ignorance-gap / EVPI upper bound on that win.

Two observations drive the rest of the paper. First, the EVPI bound is often loose: typical geometries realize only a fraction of the bound as actual adaptive advantage, and some realize essentially none. Second, the geometry $\boldsymbol{c}$ that maximizes $\mathbb{E}[IG](\boldsymbol{c})$ is generically *not* the geometry that maximizes $V_{\text{adaptive}}(\boldsymbol{c})$. In our radar case study the two objectives pick geometries whose V_{adaptive} values differ by a factor of $\sim$2.8. EVPI is a legitimate pre-solver diagnostic—if $\mathbb{E}[IG]$ is uniformly small across candidate $\boldsymbol{c}$, no adaptive policy can pay for itself—but it is not the right function to optimize $\boldsymbol{c}$ against. The right function is $V_{\text{adaptive}}(\boldsymbol{c})$, and it can only be computed by solving the adaptive problem at each candidate $\boldsymbol{c}$.

Structure of the paper. Section 2 introduces $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ and justifies its use as the reward. Sections 4–5 define the fixed design, the oracle, and the ignorance gap, and state the EVPI bound. Sections 6–7 define adaptive policies and characterize the optimal policy via Bellman's equation on the belief state. Section 8 gives the joint $(\boldsymbol{c}, \pi)$ formulation and its envelope-theorem solution. Section 9 is the radar beam-search worked example. Section 10 catalogs principled relaxations when exact Bellman is infeasible. Appendix B derives the Bellman equation from first principles.

2 The reward function: pointwise expected information gain

2.1 Setup

Let $\boldsymbol{x} \in \mathcal{X}$ be an unknown state drawn from a prior $p_0(\boldsymbol{x})$. A sensor produces observations $\boldsymbol{y}$ via a likelihood $p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, where $\boldsymbol{s} \in \mathbb{R}^{d_s}$ is the (possibly multi-step) sensor control and $\boldsymbol{c} \in \mathbb{R}^{d_c}$ is the device geometry held fixed across one sensing episode. The posterior after observing $\boldsymbol{y}$ under $(\boldsymbol{s}, \boldsymbol{c})$ is

$$p(\boldsymbol{x} \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \frac{p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})\, p_0(\boldsymbol{x})}{p(\boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})}, \qquad p(\boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c}) \;=\; \int p(\boldsymbol{y} \mid \boldsymbol{x}', \boldsymbol{s}, \boldsymbol{c})\, p_0(\boldsymbol{x}')\, d\boldsymbol{x}'. \tag{1}$$

Write $H(p) = -\int p(\boldsymbol{x}) \ln p(\boldsymbol{x})\, d\boldsymbol{x}$ for the differential entropy of a density p (nats).

2.2 Definition

Definition 1 (Pointwise expected information gain)**.** *For a fixed $\boldsymbol{x} \in \mathcal{X}$, the* pointwise expected information gain *at $\boldsymbol{x}$ under sensor control $\boldsymbol{s}$ and geometry $\boldsymbol{c}$ is*

$$\boxed{\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; H(p_0) \;-\; \mathbb{E}_{\boldsymbol{y}\sim p(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c})}\big[\, H\big(p(\,\cdot \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c})\big)\,\big].} \tag{2}$$

In words: $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is the expected decrease in posterior entropy about $\boldsymbol{x}$ when the data are generated under the true-$\boldsymbol{x}$ model. It conditions on $\boldsymbol{x}$—it is the information gain that results from an observation generated at $\boldsymbol{x}$, averaged over the randomness of the observation but not over $\boldsymbol{x}$ itself.

Equivalently (and sometimes more useful computationally), $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is the expected KL divergence from the posterior to the prior under the true-$\boldsymbol{x}$ data-generating distribution:

$$\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{y}\sim p(\boldsymbol{y}|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c})}\big[\, \mathrm{KL}\big(p(\,\cdot \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c}) \,\big\|\, p_0\big)\,\big]. \tag{3}$$

The equivalence of (2) and (3) follows from $\mathrm{KL}(p \,\|\, p_0) = -H(p) + H(p_0) +$(cross-entropy terms that integrate to see Appendix A.

2.3 Relation to standard mutual information

Standard mutual information between $\boldsymbol{x}$ and $\boldsymbol{y}$ under $(\boldsymbol{s}, \boldsymbol{c})$ is a single scalar obtained by averaging over the joint distribution:

$$I(\boldsymbol{x}; \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c}) \;=\; H(p_0) - \mathbb{E}_{\boldsymbol{y}\sim p(\cdot|\boldsymbol{s},\boldsymbol{c})}\big[H\big(p(\,\cdot \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c})\big)\big] \;=\; \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}\left[\ln \frac{p(\boldsymbol{y} \mid \boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})}{p(\boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})}\right]. \tag{4}$$

Mutual information is not a function of $\boldsymbol{x}$.

Theorem 1 (Prior average of Φ equals MI)**.**

$$\mathbb{E}_{\boldsymbol{x}\sim p_0}\big[\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})\big] \;=\; I(\boldsymbol{x}; \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c}). \tag{5}$$

Proof. Apply $\mathbb{E}_{\boldsymbol{x}}$ to (2) and use $\mathbb{E}_{\boldsymbol{x}}\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}}[f(\boldsymbol{y})] = \mathbb{E}_{\boldsymbol{y}}[f(\boldsymbol{y})]$ where the last expectation is under the marginal $p(\boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})$. The prior entropy $H(p_0)$ is constant. □

So $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is exactly the *pointwise building block whose prior-average is mutual information.* Two uses in the rest of the paper:

- Oracle / ignorance gap analyses hold $\boldsymbol{x}$ fixed; the right quantity there is $\Phi(\boldsymbol{x}, \cdot, \cdot)$ (Sections 5).
- Fixed and adaptive designs integrate over $\boldsymbol{x}$; the right quantity there is $\mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \cdot, \cdot)]$, which by Theorem 1 is MI (Sections 4, 6–7).

2.4 Non-negativity and basic properties

Remark 1 (Non-negativity in expectation)**.** *By Jensen's inequality and $\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}}[H(p(\,\cdot \mid \boldsymbol{y}))] = H(p_0) - \Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, one has $\mathbb{E}_{\boldsymbol{x}}[\Phi] = I \geq 0$. Pointwise $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ can in principle be negative for a specific $\boldsymbol{x}$ (the data generated at an "unusual" $\boldsymbol{x}$ may, in expectation, produce a posterior that is more diffuse than the prior in entropy terms), but the prior-weighted average is non-negative. In the regular, well-identifiable models used throughout this paper, pointwise $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \geq 0$ for all $\boldsymbol{x}$; negative values almost always indicate a bug or a genuinely pathological likelihood.*

Remark 2 (Multi-step Φ)**.** *For N conditionally independent measurements under a sequence $\boldsymbol{s}_{1:N}$,*

$$\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}) \;=\; H(p_0) \;-\; \mathbb{E}_{\boldsymbol{y}_{1:N}|\boldsymbol{x},\boldsymbol{s}_{1:N},\boldsymbol{c}}\big[\, H\big(p(\,\cdot \mid \boldsymbol{y}_{1:N}, \boldsymbol{s}_{1:N}, \boldsymbol{c})\big)\,\big], \tag{6}$$

with the posterior updated by the full data $\boldsymbol{y}_{1:N}$. All arguments below are phrased for the N-step reward.

2.5 Gaussian-posterior / Fisher specialization

For readers coming from classical experimental-design literature, it is useful to note the specialization of Φ in the Gaussian-posterior / Bernstein–von Mises regime. Suppose the observations are sufficiently informative that, conditional on the true $\boldsymbol{x}_0$, the posterior $p(\cdot \mid \boldsymbol{y}_{1:N})$ concentrates around $\boldsymbol{x}_0$ and is approximately Gaussian with covariance $\boldsymbol{J}_N(\boldsymbol{x}_0, \boldsymbol{s}_{1:N}, \boldsymbol{c})^{-1}$, where $\boldsymbol{J}_N$ is the usual Fisher information matrix accumulated over the N measurements. Under this approximation, $H(\text{posterior}) \approx \frac{1}{2} \ln \det(2\pi e \boldsymbol{J}_N^{-1})$, so

$$\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}) \;\approx\; \text{const} \;+\; \tfrac{1}{2} \ln \det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}). \tag{7}$$

That is, in the Gaussian / high-SNR regime, Φ specializes to the log-volume of the Fisher information ellipsoid—the classical *D-optimal* criterion of experimental design. All subsequent developments in this paper (oracle, ignorance gap, Bellman DP, joint optimization) are phrased in terms of Φ and therefore include (7) as a special case. Outside the Gaussian regime—multimodal or heavy-tailed posteriors, strong prior regularization—one must use Φ directly rather than its Fisher surrogate; the approximation (7) breaks down.

Throughout the main text we work with Φ directly. When a case study permits the Gaussian specialization (Section 10, path (2c)), we flag it explicitly.

2.6 Why Φ is the right reward

- **Information content is what a sensor produces.** Φ is a property of the joint distribution $p(\boldsymbol{x}, \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})$ induced by the sensor model. No estimator is involved in defining or computing it. It is the quantity that determines—in aggregate and through the posterior—what any downstream estimator can achieve.
- **It supports the oracle / ignorance-gap decomposition.** The pointwise form is essential: $V_{\text{oracle}}(\boldsymbol{x})$ and $IG(\boldsymbol{x})$ only make sense as functions of $\boldsymbol{x}$. Standard MI, which is a single scalar, cannot play this role.
- **It reduces to classical D-optimality in the Gaussian/BvM regime.** Equation (7) shows the framework is backward-compatible with decades of experimental-design literature, without being dependent on it.
- **Estimator-agnostic.** Debates about Cramér–Rao bounds, posterior covariance, James–Stein-type shrinkage, and biased-vs-unbiased estimators are all downstream of Φ. Maximizing Φ concentrates the posterior; how a particular estimator extracts a point from that posterior is an orthogonal question.

3 Why information gain and not PCRB? A structural asymmetry

A natural alternative to the framework just laid out would build on the prior-averaged posterior Cramér–Rao bound (PCRB) rather than on the pointwise information gain Φ. Both formulations admit a clean pre-committed design problem. The rest of the machinery—oracle, ignorance gap, Bellman reward at a realized trajectory—looks similar in structure. This section shows that the similarity is superficial: only the Φ framework admits a rigorous pointwise extension. The PCRB-based alternative is strictly weaker, by two independent mechanisms. The conclusion is not that PCRB is wrong (it is a correct and useful bound), but that it is insufficient as the foundational object of a joint geometry–policy theory.

3.1 The PCRB-based fixed-design alternative

For definiteness, consider a static unknown state $\boldsymbol{x} \sim p_0$ observed through N conditionally independent measurements $\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k$, $\boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$, with Jacobian $\boldsymbol{F}_k(\boldsymbol{x}) =$

$\partial \boldsymbol{h}/\partial \boldsymbol{x}$. Accumulating Fisher information across measurements and averaging over the prior produces the Bayesian Information Matrix

$$\boldsymbol{J}_P(\boldsymbol{s}_{1:N}, \boldsymbol{c}) \;=\; \boldsymbol{J}_0 \;+\; \sigma^{-2} \sum_{k=1}^{N} \mathbb{E}_{\boldsymbol{x}\sim p_0}\big[\boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})^\top \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\big], \tag{8}$$

where $\boldsymbol{J}_0 = -\mathbb{E}_{\boldsymbol{x}}[\nabla^2 \ln p_0(\boldsymbol{x})]$ is the prior information. The PCRB (Van Trees) states that for any estimator $\hat{\boldsymbol{x}}(\boldsymbol{y}_{1:N})$,

$$\mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}\big[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^\top\big] \;\succeq\; \boldsymbol{J}_P(\boldsymbol{s}_{1:N}, \boldsymbol{c})^{-1}. \tag{9}$$

Minimizing the log-volume of the PCRB ellipsoid gives the Bayesian D-optimal fixed-design problem

$$V_{\text{fixed}}^{\text{PCRB}}(\boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_P(\boldsymbol{s}_{1:N}, \boldsymbol{c}). \tag{10}$$

This is a clean single-level, estimator-free objective—structurally parallel to the Φ-based $V_{\text{fixed}}(\boldsymbol{c}) = \max_{\boldsymbol{s}} \mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})]$ of the next section.

If the framework stopped at the fixed-design problem, the two would be roughly interchangeable (the Φ formulation reduces to (10) in the Gaussian-posterior regime per (7)). The divergence begins when one attempts to extend the framework to a pointwise oracle, a pointwise ignorance gap, or a Bellman terminal reward.

3.2 Pointwise values can beat the aggregate in both frameworks

A tempting way to distinguish the two frameworks is to observe that pointwise values at specific $\boldsymbol{x}$ can exceed the global prior-averaged bound. This is *not* a distinguishing feature—it happens in both frameworks:

- In the Φ framework, $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is the expected information gain *conditional on* $\boldsymbol{x}$. By Theorem 1, $\mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})] = I(\boldsymbol{x}; \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})$. Pointwise $\Phi(\boldsymbol{x})$ can exceed or fall short of I; the inequality holds only in the prior-weighted integral.
- In the PCRB framework, the pointwise MSE of a Bayesian estimator (posterior mean, MAP) at a specific $\boldsymbol{x}_0$ can be *below* $\operatorname{tr} \boldsymbol{J}_P^{-1}$, at the cost of being above $\operatorname{tr} \boldsymbol{J}_P^{-1}$ at other $\boldsymbol{x}$ values. The prior acts as shrinkage: it reduces variance at favorable $\boldsymbol{x}$ (e.g., interior of the prior support) while biasing estimates at unfavorable $\boldsymbol{x}$ (e.g., boundary). The PCRB inequality (9) holds only for the prior-weighted integral.

The question is not whether pointwise values can beat aggregate bounds (yes in both), but whether each framework supplies a *rigorous pointwise object* that can serve as the building block for an oracle or ignorance-gap analysis.

3.3 The asymmetry: Φ has a pointwise summand, PCRB does not

In the Φ framework, $\Phi(\boldsymbol{x})$ is literally the pointwise integrand of MI. By Theorem 1,

$$I(\boldsymbol{x}; \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c}) \;=\; \int \Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})\, p_0(\boldsymbol{x})\, d\boldsymbol{x}. \tag{11}$$

$\Phi(\boldsymbol{x})$ is the rigorous pointwise object whose prior-weighted integral is the aggregate. The construction is exact at the level of joint distributions $p(\boldsymbol{x}, \boldsymbol{y} \mid \boldsymbol{s}, \boldsymbol{c})$: no estimator is invoked, no approximation is made, and $\Phi(\boldsymbol{x})$ is well-defined for arbitrary (possibly highly non-Gaussian) likelihoods. It supports unambiguously the oracle $V_{\text{oracle}}(\boldsymbol{x}; \boldsymbol{c}) = \max_{\boldsymbol{s}} \Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, the pointwise ignorance gap $IG(\boldsymbol{x}; \boldsymbol{c})$, and the Bellman terminal reward $R(b_{N+1}) = H(p_0) - H(b_{N+1})$.

The PCRB framework has no analogous pointwise summand. Two independent obstructions preclude a direct pointwise version:

1. **PCRB itself has no pointwise integrand.** The bound (9) is formed by averaging Fisher information over the prior *before* inversion: $\boldsymbol{J}_P = \boldsymbol{J}_0 + \sum_k \mathbb{E}_{\boldsymbol{x}}[\boldsymbol{F}_k^\top \boldsymbol{F}_k]$. The natural candidate for a pointwise summand would be $\operatorname{tr} \boldsymbol{J}_N(\boldsymbol{x})^{-1}$, the inverse Fisher information at a specific $\boldsymbol{x}$. But
$$\mathbb{E}_{\boldsymbol{x}}\big[\operatorname{tr} \boldsymbol{J}_N(\boldsymbol{x})^{-1}\big] \;\neq\; \operatorname{tr} \boldsymbol{J}_P^{-1} \qquad \text{in general,} \tag{12}$$
by Jensen's inequality (the map $\boldsymbol{A} \mapsto \operatorname{tr} \boldsymbol{A}^{-1}$ is convex on the positive-definite cone). There is no decomposition of the PCRB bound into a prior-weighted integral of pointwise bounds. The object $\operatorname{tr} \boldsymbol{J}_N(\boldsymbol{x})^{-1}$, if used at all, is an *ad hoc* choice that does not recover (9) upon averaging.
2. **No pointwise MSE bound holds for biased Bayesian estimators.** Even setting aside (1) and hoping that $\operatorname{tr} \boldsymbol{J}_N(\boldsymbol{x}_0)^{-1}$ bounds pointwise MSE at $\boldsymbol{x}_0$, one runs into the classical CRB obstruction. The pointwise classical CRB
$$\operatorname{Cov}_{\boldsymbol{y}|\boldsymbol{x}_0}\big[\hat{\boldsymbol{x}}(\boldsymbol{y})\big] \;\succeq\; \boldsymbol{J}_N(\boldsymbol{x}_0)^{-1} \tag{13}$$
holds *only for unbiased estimators* of the deterministic parameter $\boldsymbol{x}_0$. Sequential Bayesian estimators—posterior mean, MAP, any Kalman-filter-like recursive estimator—are *not* unbiased: the prior regularizes them, inducing a shrinkage bias $\boldsymbol{b}(\boldsymbol{x}_0) = \mathbb{E}[\hat{\boldsymbol{x}} \mid \boldsymbol{x}_0] - \boldsymbol{x}_0 \neq \boldsymbol{0}$ that is the price of admitting the prior at all. Therefore (13) does not apply.
The biased-CRB correction does exist,
$$\operatorname{Cov}_{\boldsymbol{y}|\boldsymbol{x}_0}\big[\hat{\boldsymbol{x}}(\boldsymbol{y})\big] \;\succeq\; \big(\boldsymbol{I} + \nabla_{\boldsymbol{x}}\boldsymbol{b}(\boldsymbol{x}_0)\big)\, \boldsymbol{J}_N(\boldsymbol{x}_0)^{-1} \big(\boldsymbol{I} + \nabla_{\boldsymbol{x}}\boldsymbol{b}(\boldsymbol{x}_0)\big)^\top, \tag{14}$$
but requires the gradient $\nabla_{\boldsymbol{x}}\boldsymbol{b}$ of the estimator's bias function—a quantity available in closed form only in trivial cases. In any practical sequential-sensing problem the bias function is implicit, defined operationally by the recursive posterior-mean update, and its gradient is neither tabulated nor tractable. The biased-CRB is therefore of theoretical but not operational value.

Together. Obstruction (1) says PCRB does not provide a pointwise summand; obstruction (2) says no other pointwise MSE bound applies to the estimators that sequential Bayesian sensing actually uses. The gap cannot be closed by more careful accounting.

3.4 Asymptotic recovery is not enough

In the Bernstein–von Mises regime (high SNR, large N, smooth identifiable model), the posterior concentrates near the true $\boldsymbol{x}_0$ with covariance approximately $\boldsymbol{J}_N(\boldsymbol{x}_0)^{-1}$, and the Bayesian estimator's pointwise MSE is $\boldsymbol{J}_N(\boldsymbol{x}_0)^{-1}$ *asymptotically.* This is an equality in the limit, not a bound at finite N.

For the joint-DP framework we need a rigorous pointwise object at *all* N—oracle and ignorance-gap analyses operate precisely in the regime where the sensor budget is finite and the posterior still carries meaningful uncertainty that the policy must navigate. Asymptotic equality at infinite N does not supply the decomposition we need at the finite horizons of Sections 7–9. Φ does, at every N, without any asymptotic assumption.

3.5 What PCRB can and cannot support

The asymmetry has concrete consequences for what each framework can rigorously formulate.

PCRB rigorously supports:

- **Prior-averaged fixed design.** $V_{\text{fixed}}^{\text{PCRB}}(\boldsymbol{c}) = \max_{\boldsymbol{s}} \ln\det \boldsymbol{J}_P$ is a well-posed single-level problem.
- **Gradient-based design.** $\partial V_{\text{fixed}}^{\text{PCRB}}/\partial \boldsymbol{c}$ and $\partial V_{\text{fixed}}^{\text{PCRB}}/\partial \boldsymbol{s}_k$ are analytic in closed form.
- **Prior-averaged monotone bounds.** Inequalities like $\mathbb{E}[\text{MSE}_{\text{fixed}}] \geq \operatorname{tr} \boldsymbol{J}_P^{-1}$.

PCRB cannot rigorously support:

- **Pointwise oracle $V_{\text{oracle}}(\boldsymbol{x})$.** Would require a pointwise MSE bound; obstructions (1) and (2) above preclude it.
- **Pointwise ignorance gap $IG(\boldsymbol{x})$.** Defined as $V_{\text{oracle}}(\boldsymbol{x}) -$ (fixed value at $\boldsymbol{x}$); inherits the obstruction.
- **Bellman terminal reward at a realized trajectory.** The terminal $\mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})]$ in (32) requires pointwise evaluation of the reward along each observation path. PCRB provides no such pointwise quantity.

These are not omissions that additional work could fill in; they are structurally absent because the PCRB inequality is an *integral* bound by construction, with no pointwise decomposition surviving the prior-averaging step that defines $\boldsymbol{J}_P$.

3.6 The Φ framework subsumes the PCRB framework in the Gaussian regime

In the Gaussian-posterior / BvM regime of Section 2.5,

$$\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \;\approx\; \text{const} \;+\; \tfrac{1}{2} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}). \tag{15}$$

Taking prior averages and applying the Φ-to-MI identity:

$$V_{\text{fixed}}(\boldsymbol{c}) \;=\; \max_{\boldsymbol{s}} \; \mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})] \;\approx\; \max_{\boldsymbol{s}} \; \tfrac{1}{2}\, \mathbb{E}_{\boldsymbol{x}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})] \;+\; \text{const}. \tag{16}$$

This is closely related to (10), the PCRB-based objective, though the two differ by the order of log-det and expectation (Jensen gap). Both recover the classical Bayesian D-optimal program in the Gaussian regime, up to this reordering.

The Φ framework therefore includes the PCRB-style fixed-design objective as a special case (via (16)), extends it pointwise (via $\Phi(\boldsymbol{x}, \cdot)$), and remains rigorous outside the Gaussian regime (exact MI without any approximation). The reverse containment does not hold: PCRB admits no pointwise extension, and ceases to be a bound outside the regime where the estimators happen to be asymptotically unbiased.

3.7 PCRB as an independent MSE-anchored baseline

The structural limitations of Section 3.3 do *not* disqualify the PCRB framework from playing a useful role in the joint-design picture—they only prevent it from serving as the *foundation*. One constructive use remains: the PCRB-based joint fixed-design optimization (10) serves as an **independent, MSE-anchored reference baseline** against which the Φ-based joint $(\boldsymbol{c}, \pi)$ DP result can be reported.

The baseline is well-defined without any of the obstructions discussed above:

- The PCRB inequality (9) is rigorous at any finite N for *any* estimator satisfying the mild regularity conditions of the bound. No unbiasedness is assumed—the inequality is in prior-averaged form, which tolerates the bias inherent in Bayesian estimators because the expectation smooths over $\boldsymbol{x}$.
- The joint optimization

$$\left(\boldsymbol{c}^\star_{\text{PCRB}},\, \boldsymbol{s}^\star_{\text{PCRB}}\right) \;=\; \operatorname{argmax}_{\boldsymbol{c},\, \boldsymbol{s}_{1:N}} \; \ln\det \boldsymbol{J}_P(\boldsymbol{s}_{1:N}, \boldsymbol{c}) \tag{17}$$

is a single-level problem in $d_c + Nd_s$ real variables—flat, amenable to gradient descent, and well-studied in the classical experimental-design literature. It produces a geometry $\boldsymbol{c}^\star_{\mathrm{PCRB}}$ that rigorously minimizes the log-volume of the prior-averaged MSE ellipsoid.

- The output geometry carries a concrete MSE interpretation: any estimator deployed at $\boldsymbol{c}^\star_{\mathrm{PCRB}}$ with fixed schedule $\boldsymbol{s}^\star_{\mathrm{PCRB}}$ has prior-averaged MSE bounded below by $\operatorname{tr} \boldsymbol{J}_P(\boldsymbol{s}^\star_{\mathrm{PCRB}}, \boldsymbol{c}^\star_{\mathrm{PCRB}})^{-1}$.

This baseline has three operational uses in reporting joint-$(\boldsymbol{c}, \pi)$ DP results.

Sanity check on adaptive gain. Evaluating V_{adaptive} at $\boldsymbol{c}^\star_{\mathrm{PCRB}}$ (the baseline geometry) isolates the policy contribution: if the adaptive policy at the PCRB-optimal geometry already substantially beats the fixed $\boldsymbol{s}^\star_{\mathrm{PCRB}}$, we learn that adaptation pays off even on a geometry designed for the non-adaptive setting. If not, the case for joint-policy optimization is weak at this geometry, independent of whether a better $\boldsymbol{c}$ exists.

MSE-anchored quantitative claim. Let

$$(\boldsymbol{c}^\star_1,\, \pi^\star_1) \;=\; \operatorname{argmax}_{\boldsymbol{c},\,\pi}\; W_1(p_0;\, \boldsymbol{c}) \qquad \text{(joint } \boldsymbol{c} + \text{DP result of Section 8)} \tag{18}$$

and

$$(\boldsymbol{c}^\star_2,\, \boldsymbol{s}^{(2)\star}_{1:N}) \;=\; \operatorname{argmax}_{\boldsymbol{c},\,\boldsymbol{s}_{1:N}}\; \ln\det \boldsymbol{J}_P(\boldsymbol{s}_{1:N}, \boldsymbol{c}) \qquad \text{(PCRB-baseline result (17))}. \tag{19}$$

Denote by $\overline{\mathrm{MSE}}_1$ and $\overline{\mathrm{MSE}}_2$ the respective prior-averaged mean-square errors achieved in deployment—the first under the adaptive policy $\pi^\star_1$ at geometry $\boldsymbol{c}^\star_1$, the second under the fixed schedule $\boldsymbol{s}^{(2)\star}_{1:N}$ at geometry $\boldsymbol{c}^\star_2$. Both are concrete operational quantities measurable from Monte Carlo simulation or from deployed data.

The ratio

$$\text{MSE gain} \;=\; \frac{\overline{\mathrm{MSE}}_2}{\overline{\mathrm{MSE}}_1} \tag{20}$$

gives a reviewer-friendly, estimator-level, MSE-denominated statement of how much the joint-$(\boldsymbol{c}, \pi)$ DP formulation buys over the classical PCRB-optimal fixed design. Unlike an information-gain ratio—which invites the reviewer's "but what does this buy me in MSE?"—the PCRB-anchored ratio answers that question directly, in the same units in which the sensor's deployed performance will be reported.

Note that $\overline{\mathrm{MSE}}_1 < \overline{\mathrm{MSE}}_2$ is *not* guaranteed by theory: the PCRB baseline solves a different (prior-averaged) problem from the Bellman DP (pointwise-Bellman-expected), and the two objectives can disagree on which $\boldsymbol{c}$ is best even within the Gaussian regime where they are closest. Empirically checking that $\overline{\mathrm{MSE}}_1 < \overline{\mathrm{MSE}}_2$ is therefore a meaningful test—not a tautology—and a passing result is a quantitative endorsement of the joint-DP framework in the currency that matters for deployment.

Null hypothesis for the whole framework. If $V_{\mathrm{adaptive}}(\boldsymbol{c}^\star)$ does not exceed $V^{\mathrm{PCRB}}_{\mathrm{fixed}}(\boldsymbol{c}^\star_{\mathrm{PCRB}})$ after the Gaussian-regime translation of (16), then the joint-$(\boldsymbol{c}, \pi)$ DP machinery does not pay for itself relative to classical Bayesian D-optimal design. In that case, one should report the PCRB-optimal $\boldsymbol{c}^\star_{\mathrm{PCRB}}$ with its fixed schedule and stop. This is the quantitative analog of Corollary 1, operationalized in MSE units.

Summary of the relationship. The two frameworks are *complementary*, with distinct scopes:

- The Φ framework supplies the *internal machinery*—oracle, ignorance gap, Bellman reward, envelope-theorem gradient, joint optimization. It is rigorous pointwise, estimator-agnostic, and valid at all N and all likelihoods.

- The PCRB framework supplies the *external MSE anchor*—a rigorously prior-averaged baseline geometry $\boldsymbol{c}^\star_{\mathrm{PCRB}}$ and a concrete MSE bound against which the Φ-framework's output can be benchmarked.

Neither replaces the other. The Φ framework without the PCRB baseline lacks a concrete MSE endpoint for its information-theoretic quantities; the PCRB framework without Φ lacks the pointwise machinery needed to formulate the adaptive half of the problem. Reporting the joint-DP result and the PCRB baseline side-by-side is the natural way to close both gaps.

3.8 Summary

Capability	Φ framework	PCRB framework
Prior-averaged fixed design	yes, rigorous at any N	yes, rigorous at any N
Pointwise reward at specific $\boldsymbol{x}$	$\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ (exact integrand of MI)	ad hoc; no rigorous candidate
Valid for biased estimators	yes (estimator-agnostic)	no; bound requires unbiasedness
Oracle $V_{\mathrm{oracle}}(\boldsymbol{x})$	well-defined pointwise	not available
Ignorance gap $IG(\boldsymbol{x})$	well-defined pointwise	not available
Bellman terminal reward at b_{N+1}	$H(p_0) - H(b_{N+1})$	no pointwise analog
Gaussian/BvM specialization	recovers Bayesian D-optimal	(definitional)
Validity outside Gaussian regime	exact	bound does not apply

Three observations consolidate the asymmetry. First, both frameworks admit the Jensen-gap phenomenon in which pointwise values can exceed the aggregate bound; this is neither special nor distinguishing. Second, $\Phi(\boldsymbol{x})$ admits a rigorous pointwise interpretation as the pointwise integrand of MI, whereas no such pointwise integrand of PCRB exists. Third, even if one wished to bound pointwise MSE directly, no such non-asymptotic bound applies to the biased Bayesian estimators actually used in sequential sensing. The Φ framework is therefore the minimal rigorous setting in which the joint geometry–policy theory—oracle, ignorance gap, Bellman reward, envelope-theorem gradient, and joint optimization—can be stated at all.

4 Pre-computed sensors: the fixed-design problem

A *pre-computed* (or *fixed*, or *non-adaptive*) design commits the sensor sequence $\boldsymbol{s}_{1:N}$ at design time, before any data are collected. Each $\boldsymbol{s}_k$ is a fixed vector; none depends on $\boldsymbol{y}_{1:k-1}$.

Definition 2 (Fixed-design value)**.** *The fixed-design value at geometry $\boldsymbol{c}$ is*

$$V_{\mathrm{fixed}}(\boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_{1:N}} \mathbb{E}_{\boldsymbol{x}\sim p_0}\big[\,\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})\,\big] \;=\; \max_{\boldsymbol{s}_{1:N}} I(\boldsymbol{x};\, \boldsymbol{y}_{1:N} \mid \boldsymbol{s}_{1:N}, \boldsymbol{c}), \tag{21}$$

with optimizer $\boldsymbol{s}^\star_{\mathrm{fixed}}(\boldsymbol{c}) = \mathrm{argmax}_{\boldsymbol{s}_{1:N}} \mathbb{E}_{\boldsymbol{x}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})]$.

The equality with mutual information on the right is Theorem 1; the fixed-design problem is a single-level maximization of $I(\boldsymbol{x}; \boldsymbol{y}_{1:N})$ over the Nd_s-dimensional vector of sensor parameters (plus any geometry $\boldsymbol{c}$ one includes in the outer max).

Optimization structure. The fixed-design problem is a *single-level* optimization—no inner argmax inside an outer max. If Φ is smooth in $(\boldsymbol{s}, \boldsymbol{c})$, gradient-based optimization applies directly:

$$\nabla_{\boldsymbol{s}_k} V_{\mathrm{fixed}}(\boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x}}\big[\nabla_{\boldsymbol{s}_k}\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})\big]_{\boldsymbol{s}^\star}, \qquad \nabla_{\boldsymbol{c}} V_{\mathrm{fixed}}(\boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x}}\big[\nabla_{\boldsymbol{c}}\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})\big]_{\boldsymbol{s}^\star}. \tag{22}$$

Monte Carlo or grid-based estimation of $\mathbb{E}_{\boldsymbol{x}}[\cdot]$ typically suffices.

Historical note. In the Gaussian-posterior specialization (7), (21) reduces to the classical Bayesian D-optimal design problem,

$$V_{\text{fixed}}^{\text{Fisher}}(\boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_{1:N}} \; \mathbb{E}_{\boldsymbol{x}}\big[\, \tfrac{1}{2} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}) \,\big] \;+\; \text{const}, \tag{23}$$

recovering a century of sequential-design literature. We keep (21) as the primary statement because it is valid beyond the Gaussian regime.

5 The ignorance gap

The fixed design (21) hedges against the unknown $\boldsymbol{x}$ by averaging over the prior. How much worse is this hedging than an oracle that *sees* $\boldsymbol{x}$ and picks sensors specifically for it? The ignorance gap quantifies this pointwise.

5.1 Oracle value and ignorance gap

Definition 3 (Oracle value)**.** *The oracle value at state* $\boldsymbol{x}$ *is*

$$V_{\text{oracle}}(\boldsymbol{x};\, \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_{1:N}} \; \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}). \tag{24}$$

This is the best expected information gain at $\boldsymbol{x}$ if the designer were allowed to see $\boldsymbol{x}$ and pick sensors specifically for it. It is pointwise in $\boldsymbol{x}$. No realizable policy (adaptive or otherwise) can exceed it, because no realizable policy has access to $\boldsymbol{x}$ before data collection.

Definition 4 (Ignorance gap)**.** *The pointwise* ignorance gap *at* $\boldsymbol{x}$ *is*

$$\boxed{IG(\boldsymbol{x}; \boldsymbol{c}) \;=\; V_{\text{oracle}}(\boldsymbol{x}; \boldsymbol{c}) \;-\; \Phi\big(\boldsymbol{x},\, \boldsymbol{s}^{\star}_{\text{fixed}}(\boldsymbol{c}),\, \boldsymbol{c}\big) \;\geq\; 0.} \tag{25}$$

$IG(\boldsymbol{x})$ measures how much better the sensors *could* be at $\boldsymbol{x}$ if tailored specifically to it, relative to the one-size-fits-all $\boldsymbol{s}^{\star}_{\text{fixed}}$. The prior-expected ignorance gap is

$$\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x}; \boldsymbol{c})] \;=\; \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x}; \boldsymbol{c})] \;-\; V_{\text{fixed}}(\boldsymbol{c}). \tag{26}$$

5.2 The EVPI bound

Introducing (a preview of Section 6) the value of the optimal adaptive policy $V_{\text{adaptive}}(\boldsymbol{c})$:

Theorem 2 (EVPI bound)**.**

$$\boxed{0 \;\leq\; V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c}) \;\leq\; \mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x}; \boldsymbol{c})].} \tag{27}$$

Proof. Lower bound: a fixed schedule is a (constant) adaptive policy, so $V_{\text{adaptive}} \geq V_{\text{fixed}}$. Upper bound: no policy has access to $\boldsymbol{x}$, so $V_{\text{adaptive}} \leq \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x})]$; subtract V_{fixed} from both sides and apply (26). □

This is the *Expected Value of Perfect Information* (EVPI) bound from decision theory, specialized to information-gain rewards.

Corollary 1. *If* $\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x}; \boldsymbol{c})] \approx 0$*, then no adaptive policy can significantly outperform the pre-computed design at that* $\boldsymbol{c}$*.*

Remark 3 (EVPI is necessary but not sufficient for adaptive gain)**.** *The converse of Corollary 1 does* not *hold. A large* $\mathbb{E}[IG]$ *indicates that adaptation* might *help but does not guarantee it. The inequality* (27) *is generically loose, and the realized gap* $V_{\text{adaptive}} - V_{\text{fixed}}$ *depends sensitively on the geometry* $\boldsymbol{c}$*. Two designs with the same* $\mathbb{E}[IG]$ *can have radically different* $V_{\text{adaptive}} - V_{\text{fixed}}$*; a design with smaller* $\mathbb{E}[IG]$ *can have a* larger *realized adaptive advantage than one with larger* $\mathbb{E}[IG]$*. Section 8.4 quantifies this in the radar case study, where the EVPI-maximizing geometry has* $\sim 2.8\times$ smaller V_{adaptive} *than the joint-DP optimum within the same beam family. Consequently,* $\mathbb{E}[IG]$ *is a pre-solver diagnostic, not a design target: it says* whether *to engage the adaptive loop, not* where *the joint* $(\boldsymbol{c}, \pi)$ *optimum lies.*

5.3 When is the ignorance gap small?

$IG(\boldsymbol{x}) \equiv 0$ exactly when $\operatorname{argmax}_{\boldsymbol{s}_{1:N}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ does not depend on $\boldsymbol{x}$—i.e., the same sensor sequence is oracle-optimal regardless of the true state. This holds for *translation-covariant* likelihoods where $\boldsymbol{x}$ shifts the data but does not change which sensor configuration extracts maximum information (e.g., estimating the mean of a scalar under Gaussian noise with a fixed noise variance). More generally, $IG(\boldsymbol{x})$ is small when the oracle map $\boldsymbol{x} \mapsto \boldsymbol{s}^\star(\boldsymbol{x})$ is nearly constant over the prior support—typically when the likelihood's $\boldsymbol{x}$-dependence is weak relative to its $\boldsymbol{s}$-dependence.

5.4 Computability of the ignorance gap

The ignorance gap is computable from the fixed-design solution alone—no adaptive solver required:

1. Solve (21) to obtain $\boldsymbol{s}^\star_{\text{fixed}}(\boldsymbol{c})$.
2. For a Monte Carlo sample $\{\boldsymbol{x}^{(i)}\}_{i=1}^M \sim p_0$:
   (a) Evaluate $\Phi(\boldsymbol{x}^{(i)}, \boldsymbol{s}^\star_{\text{fixed}}, \boldsymbol{c})$.
   (b) Solve $V_{\text{oracle}}(\boldsymbol{x}^{(i)}; \boldsymbol{c}) = \max_{\boldsymbol{s}_{1:N}} \Phi(\boldsymbol{x}^{(i)}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ (an Nd_s-variable optimization).
   (c) Compute $IG(\boldsymbol{x}^{(i)})$ as their difference.
3. Report $\widehat{\mathbb{E}}[IG]$, $\widehat{\text{std}}[IG]$, $\widehat{\max}[IG]$, $\Pr[IG > \epsilon]$.

If $\widehat{\mathbb{E}}[IG] \ll$ (prior entropy), Corollary 1 says the fixed design is essentially optimal. This offline check costs M small optimizations and should precede any investment in an adaptive solver.

Diagnostic statistics. The distribution of $IG(\boldsymbol{x})$ carries more information than its mean:

Statistic	Interpretation
$\mathbb{E}[IG]$	Upper bound on adaptation value (EVPI, in nats)
$\text{std}[IG]/\mathbb{E}[IG]$	Heterogeneity: large $\Rightarrow$ benefit concentrated on certain $\boldsymbol{x}$
$\max IG$	Worst-case suboptimality of fixed design
$\Pr(IG > \epsilon)$	Fraction of state space where adaptation matters

6 Adaptive policies

When $\mathbb{E}[IG]$ is not negligible, the fixed design leaves value on the table: by observing intermediate $\boldsymbol{y}_k$ and using them to select later $\boldsymbol{s}_{k+1}$ we may extract information the fixed schedule misses. This motivates *adaptive* policies.

6.1 Definition

An **adaptive policy** is a sequence of measurable functions

$$\pi = \{\pi_1, \pi_2, \ldots, \pi_N\}, \qquad \pi_k : \mathcal{Y}^{k-1} \to \mathbb{R}^{d_s}, \tag{28}$$

where $\pi_k(\boldsymbol{y}_{1:k-1})$ is the sensor action at step k, a function of the data observed *before* step k. By convention π_1 takes no argument.

Equivalently, the history $\boldsymbol{y}_{1:k-1}$ enters only through the posterior belief

$$b_k(\boldsymbol{x}) \;=\; p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1}, \boldsymbol{s}_{1:k-1}, \boldsymbol{c}), \tag{29}$$

so we may write $\pi_k : b_k \mapsto \boldsymbol{s}_k$—a map from beliefs to sensor parameters. The belief b_k is the *sufficient statistic* for all future decisions.

6.2 Contrast with pre-computed sensors

Property	Pre-computed (§4)	Adaptive (this section)
$\boldsymbol{s}_k$ is	a fixed vector	a function $\pi_k(\boldsymbol{y}_{1:k-1}) = \pi_k(b_k)$
Chosen	before any data	after each observation
Depends on $\boldsymbol{y}_{1:k-1}$?	No	Yes
Search space	$\mathbb{R}^{Nd_s}$ (finite-dim)	function space (infinite-dim)
Benefit	hedges average-case $\boldsymbol{x}$	can target inferred $\hat{\boldsymbol{x}}$
Value	V_{fixed} (21)	$V_{\text{adaptive}} \le V_{\text{fixed}} + \mathbb{E}[IG]$

Remark 4. *An "adaptive" policy is a* function *of the data, not merely a computation performed at run time. A schedule whose numerical values are computed during data collection but whose* functional form *does not use $\boldsymbol{y}_{1:k-1}$ is still a fixed design in this sense.*

7 Finding the optimal adaptive policy: Bellman's equation

The adaptive problem

$$V_{\text{adaptive}}(\boldsymbol{c}) \;=\; \max_{\pi}\; \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}\big[\,\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}), \boldsymbol{c})\,\big] \tag{30}$$

is a maximization over functions and hence intractable at face value. The Bellman recursion reduces it to a sequence of single-step maximizations.

7.1 Value-to-go

Define the *value-to-go* $W_k(b_k; \boldsymbol{c})$: the best expected remaining information gain from step k onward, starting at belief b_k. Boundary values:

- $W_1(p_0; \boldsymbol{c}) = V_{\text{adaptive}}(\boldsymbol{c})$—the quantity of interest.
- $W_{N+1}(b_{N+1}; \boldsymbol{c}) = H(p_0) - H(b_{N+1})$—the accumulated information at the terminal belief.

Writing the terminal this way lets W_{N+1} encode the remaining value-to-go when no decisions remain, and W_1 equal $\mathbb{E}[\Phi]$ at the prior under the optimal policy.

7.2 Bellman recursion

Theorem 3 (Bellman recursion; see Appendix B for derivation)**.**

$$W_k(b_k; \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k}\; \mathbb{E}_{\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k, \boldsymbol{c}}\big[\,W_{k+1}(b_{k+1}; \boldsymbol{c})\,\big], \quad k = N, \ldots, 1, \tag{31}$$

$$W_{N+1}(b_{N+1}; \boldsymbol{c}) \;=\; H(p_0) - H(b_{N+1}), \tag{32}$$

where b_{k+1} is the Bayesian posterior update of b_k given $(\boldsymbol{s}_k, \boldsymbol{y}_k)$. The optimal policy is

$$\pi_k^\star(b_k; \boldsymbol{c}) \;=\; \operatorname{argmax}_{\boldsymbol{s}_k}\; \mathbb{E}_{\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k, \boldsymbol{c}}\big[\,W_{k+1}(b_{k+1}; \boldsymbol{c})\,\big], \tag{33}$$

and $V_{\text{adaptive}}(\boldsymbol{c}) = W_1(p_0; \boldsymbol{c})$.

The equations (31)–(32) reduce an infinite-dimensional optimization to a sequence of finite-dimensional $\max_{\boldsymbol{s}_k}$'s. The price is that W_k is a function on belief space, which is itself infinite-dimensional in general. Exact representation is possible in three cases: (i) *finite discrete states*, where b_k lies in a simplex and W_k is piecewise-linear convex (Sondik); (ii) *gridded continuous states with discrete actions and countable observations*, where the reachable belief tree is finite and W_k is tabulated at each reachable b_k; (iii) *linear-Gaussian problems*, where b_k is parameterized by its mean and covariance. Outside these, approximations are required (Section 10).

7.3 Oracle value-to-go and conditional ignorance gap

Analogous to the unconditional ignorance gap, define

$$V_k^\star(\boldsymbol{x};\boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_{k:N}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:k-1}, \boldsymbol{s}_{k:N}, \boldsymbol{c}) \tag{34}$$

(oracle value-to-go at step k given committed $\boldsymbol{s}_{1:k-1}$). Since no policy beats the oracle, $W_k(b_k;\boldsymbol{c}) \le \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x};\boldsymbol{c})]$, and the *conditional ignorance gap*

$$\Gamma_k(b_k;\boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x};\boldsymbol{c})] - W_k(b_k;\boldsymbol{c}) \;\ge\; 0 \tag{35}$$

satisfies $\Gamma_1(p_0;\boldsymbol{c}) = \mathbb{E}[V_{\text{oracle}}] - V_{\text{adaptive}}$. Substituting $W_k = \mathbb{E}[V_k^\star] - \Gamma_k$ into (31) yields a DP on Γ_k with an explicit exploration–exploitation structure; this *gap DP* is useful for theoretical analysis and is standard in the literature but is not required here.

8 Joint optimization of geometry and adaptive policy

The full joint problem over the device geometry $\boldsymbol{c}$ and the adaptive policy π is

$$\boxed{\max_{\boldsymbol{c}}\; W_1(p_0;\,\boldsymbol{c}) \;=\; \max_{\boldsymbol{c}} \max_{\pi} \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}\big[\,\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}), \boldsymbol{c})\,\big].} \tag{36}$$

The geometry enters every level: the likelihoods $p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})$, the posterior-update operator, the terminal $H(b_{N+1})$ via the induced belief trajectory, and hence the oracle schedule $\boldsymbol{s}^\star(\boldsymbol{x};\boldsymbol{c})$ and the Bellman-optimal policy $\pi^\star(\boldsymbol{c})$.

8.1 Envelope theorem for the outer gradient

Naive differentiation of (36) would require $\partial\pi^\star/\partial\boldsymbol{c}$, a formidable object that is ill-defined at policy-switch boundaries. The envelope theorem (Danskin) says we don't need it:

$$\boxed{\left.\frac{dW_1}{d\boldsymbol{c}}\right|_{\boldsymbol{c}=\boldsymbol{c}_0} \;=\; \frac{\partial}{\partial\boldsymbol{c}}\,\mathbb{E}\big[\,\Phi\big(\boldsymbol{x},\,\pi^\star(\boldsymbol{y};\,\boldsymbol{c}_0),\,\boldsymbol{c}\big)\,\big]\Big|_{\boldsymbol{c}=\boldsymbol{c}_0}.} \tag{37}$$

In words: at the optimal policy for $\boldsymbol{c}_0$, the first-order effect of perturbing the policy vanishes (by the first-order optimality condition on π). We differentiate Φ as if the policy were fixed—only the direct effect of $\boldsymbol{c}$ on the likelihood and the terminal entropy matter, evaluated at whatever sensor values $\pi^\star(\boldsymbol{c}_0)$ happens to emit.

8.2 Policy iteration

Equation (37) enables an alternating policy-iteration scheme:

1. **Policy step.** For current $\boldsymbol{c}_t$, solve (or approximate) the Bellman recursion (31) to obtain $\pi^\star(\boldsymbol{c}_t)$.
2. **Geometry step.** Take a gradient step on $\boldsymbol{c}$ using $\partial\Phi/\partial\boldsymbol{c}$ evaluated at $\pi^\star(\boldsymbol{c}_t)$ fixed (the RHS of (37)).

3. Repeat until convergence.

Step 1 contains the computational burden; step 2 is the same adjoint/sensitivity computation one uses in the fixed-design problem, just with the sensors coming from $\pi^\star$ rather than a constant schedule.

8.3 Reduction by regime

The ignorance gap $\mathbb{E}[IG]$ controls how much of this machinery is needed:

Regime	Joint problem reduces to	Solver
$\mathbb{E}[IG] \approx 0$	Fixed design: $\max_{\boldsymbol{c},\boldsymbol{s}_{1:N}} \mathbb{E}_{\boldsymbol{x}}[\Phi]$	Single-level gradient descent
$\mathbb{E}[IG]$ moderate	Certainty-equivalent policy iteration	Oracle map + filter + adjoint
$\mathbb{E}[IG]$ large	Full Bellman DP + policy iteration	Exact DP or parameterized approximation

The computability check of Section 5 decides which row applies before any solver is implemented.

8.4 The added value of joint optimization

The decoupled program

"*Pick $\boldsymbol{c}$ to maximize $\mathbb{E}[IG](\boldsymbol{c})$ (since large EVPI promises adaptive benefit). Separately, at run time, execute an adaptive policy.*"

is *not* equivalent to the joint optimization (36). Three facts together make joint optimization a distinct and necessary step:

1. $\mathbb{E}[IG](\boldsymbol{c})$ and $V_{\text{adaptive}}(\boldsymbol{c}) - V_{\text{fixed}}(\boldsymbol{c})$ are *different functions of* $\boldsymbol{c}$. A $\boldsymbol{c}$ maximizing one need not maximize the other.
2. The ratio $\big(V_{\text{adaptive}} - V_{\text{fixed}}\big)/\mathbb{E}[IG]$ (the *EVPI saturation fraction*) varies substantially with $\boldsymbol{c}$ and is typically well below 1.
3. The $\boldsymbol{c}$ maximizing V_{adaptive} generally differs from the $\boldsymbol{c}$ maximizing V_{fixed}.

Radar case study, within-Gaussian-family comparison (see Section 9). For the $K = 16$, $N = 4$ benchmark at $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$, restricting attention to the Gaussian-beam family $G_w(\alpha) = \exp(-\alpha^2/(2w^2))$ and comparing the two extremes against the in-family optimum $w^\star = 0.60$ rad:

geometry $\boldsymbol{c}$	$\mathbb{E}[IG]$ (EVPI)	V_{adaptive}	realized	saturation
Gaussian $w = 0.05$ (small-w extreme)	2.08 nats	0.62 nats	0.12 nats	5.8%
Gaussian $w = \pi$ (large-w extreme)	0.09 nats	0.12 nats	0.002 nats	2.4%
Gaussian $w = 0.60$ (in-family $w^\star$)	0.64 nats	**0.87** nats	0.11 nats	**17.0**%

The small-w extreme has the *largest* $\mathbb{E}[IG]$ (by a factor of ~ 3 over $w^\star$), yet only the second-largest V_{adaptive} and a saturation of just 5.8%. The large-w extreme has both the smallest $\mathbb{E}[IG]$ and the smallest V_{adaptive}. The in-family optimum $w = 0.60$ *dominates* both extremes on V_{adaptive} despite having neither the largest $\mathbb{E}[IG]$ nor the largest realized gain. Routing the geometry choice through EVPI alone would have picked the small-w design; the joint DP picks $w = 0.60$.

Practical prescription.

1. Compute $\mathbb{E}[IG]$ at a few candidate $\boldsymbol{c}$. If uniformly $\ll$ prior entropy, (21) suffices (Corollary 1).

2. Otherwise run the joint $(\boldsymbol{c}, \pi)$ optimization (36) via policy iteration with envelope-theorem geometry gradients. The EVPI diagnostic determines *whether* to engage this loop; the joint DP determines *where* its optimum lies.

EVPI and joint-DP serve complementary but non-interchangeable roles.

9 A tractable case study: adaptive radar beam search

We conclude with a concrete problem that exercises every ingredient of the framework: $\mathbb{E}[IG]$ is demonstrably large, the geometry $\boldsymbol{c}$ exerts first-order control on achievable performance, and the Bellman recursion (31) is exactly solvable. The policy-iteration scheme of Section 8 becomes a concrete algorithm. The problem is deliberately small and canonical so that every claim can be verified numerically.

9.1 Problem setup

State. A single target occupies one of K discrete cells at distinct angular positions. We use $K = 16$ cells equally spaced on a circular arc at fixed range from the antenna, so cell $z \in \{1, \ldots, K\}$ has bearing $\varphi(z) = 2\pi(z-1)/K$. The prior on z is uniform.

Actions. At each step $k = 1, \ldots, N$ (we use $N = 4$), the operator steers the antenna to a pointing direction $\theta_k \in \Theta = \{\theta_1, \ldots, \theta_M\}$ taken from a finite set of candidate pointings (we take $M = K$).

Observations. Each pointing returns a binary detection $y_k \in \{0, 1\}$ with Bernoulli likelihood

$$\Pr(y_k = 1 \mid z, \theta_k; \boldsymbol{c}) \;=\; p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}})\, G_{\boldsymbol{c}}\big(\theta_k - \varphi(z)\big), \tag{38}$$

where $G_{\boldsymbol{c}}(\alpha) \in [0, 1]$ is the normalized antenna gain pattern at angular offset α from the main-lobe direction.

Geometry. The design vector $\boldsymbol{c}$ is the full set of antenna-array parameters: a phased array of N_e elements at positions $\{\boldsymbol{r}_n\}$ driven by complex excitations $\{a_n\}$. The far-field array factor is

$$F_{\boldsymbol{c}}(\alpha) \;=\; \sum_{n=1}^{N_e} a_n \, \exp\!\big(\mathrm{i}\, k_0\, \boldsymbol{r}_n \cdot \hat{\boldsymbol{\alpha}}\big), \qquad k_0 = 2\pi/\lambda, \tag{39}$$

and $G_{\boldsymbol{c}}(\alpha) = |F_{\boldsymbol{c}}(\alpha)|^2 / \max_{\alpha'} |F_{\boldsymbol{c}}(\alpha')|^2$. The design vector is $\boldsymbol{c} = (\{\boldsymbol{r}_n\}, \{a_n\})$ with real dimension $d_c \sim 2$–$4N_e$ depending on what is free. Typical $N_e \in \{8, \ldots, 64\}$ gives d_c in the tens to low hundreds.

Reward. For this discrete problem the reward is

$$\Phi(z, \theta_{1:N}, \boldsymbol{c}) \;=\; \ln K - H(z \mid y_{1:N}, \theta_{1:N}, \boldsymbol{c}) \tag{40}$$

—the discrete specialization of Definition 1 with $H(p_0) = \ln K$ (uniform prior over K cells). Thus $\Phi \in [0, \ln K]$, with $\Phi = \ln K$ at full resolution.

9.2 The ignorance gap is large

Oracle. Knowing z, the oracle always points $\theta_k^\star = \varphi(z)$ (or the nearest element of Θ), receiving an on-target Bernoulli with probability p_{D} every step. For $p_{\mathrm{D}} \gg p_{\mathrm{FA}}$ and $N \geq \log_2 K$ (saturated at $K = 16$, $N = 4$), the posterior is essentially fully resolved: $V_{\mathrm{oracle}}(z) \approx \ln K$.

Fixed design (narrow-beam family). Assume geometry $\boldsymbol{c}$ produces a single-main-lobe, directional gain with main-lobe width $\Delta\theta_{\mathrm{ML}}(\boldsymbol{c}) \lesssim 2\pi/K$ (one cell per main lobe). With $N < K$ narrow beams, at most N distinct cells are probed. Conditioning on whether the target lies inside one of those N cells (probability N/K) or outside (probability $(K-N)/K$):

$$V_{\text{fixed}} \;\approx\; \ln K - \frac{K-N}{K}\ln(K-N). \tag{41}$$

EVPI. Combining,

$$\boxed{\mathbb{E}[IG] \;=\; V_{\text{oracle}} - V_{\text{fixed}} \;\approx\; \frac{K-N}{K}\ln(K-N) \;\text{ nats,}} \tag{42}$$

which scales as $\sim \ln K$ for $K \gg N$ and vanishes as $N \to K$. At $K = 16$, $N = 4$, $\mathbb{E}[IG] \approx (12/16)\ln 12 \approx 1.86$ nats—a substantial EVPI upper bound whose size, however, does *not* imply a large adaptive advantage (cf. Remark 3).

9.3 Geometry has first-order impact

The design $\boldsymbol{c}$ shapes every feature of $G_{\boldsymbol{c}}$ independently—main-lobe width, sidelobe structure, nulls, grating lobes—and each feeds into the Bernoulli likelihood (38) in a distinct way. Main-lobe width (aperture size), first sidelobe level (amplitude taper), grating-lobe behavior (element spacing), and null placement (complex excitations) all have first-order impact on both V_{adaptive} and V_{fixed}, and they interact. The joint optimum over $(\boldsymbol{c}, \pi)$ solves a d_c-dimensional problem whose objective is modulated by the adaptive policy that will run on top.

Rough sizing. Binary observations limit single-step information to $\ln 2$ nats, so full K-cell resolution requires $N \geq \log_2 K$. For a flat "in-beam / out-of-beam" idealization with m cells per main lobe, single-measurement mutual information is maximized at $m^\star \approx K/2$, giving main-lobe width

$$\Delta\theta^\star_{\mathrm{ML}} \;\sim\; \pi \qquad \text{(half the ring).} \tag{43}$$

A key fact: N contiguous-arc beams produce at most $2N$ arc boundaries and hence $2N$ distinguishable segments of $K/(2N)$ cells each. Adaptive policies can bisect the current posterior each step—a binary search reaching single-cell resolution in $\log_2 K$ steps—while fixed designs cannot.

Two regimes at opposite extremes of the beam-width knob:

- *Narrow* ($m \approx 1$, $\Delta\theta_{\mathrm{ML}} \lesssim 2\pi/K$). Each measurement probes one cell; fixed and adaptive achieve $V_{\text{fixed}} = V_{\text{adaptive}} \approx \ln K - \frac{K-N}{K}\ln(K-N) \approx 0.91$ nats at $K = 16, N = 4$. EVPI $= 1.86$ nats; adaptive advantage $= 0$. **Large EVPI, zero realized adaptive advantage**—the canonical counter-example to the "large EVPI implies valuable adaptation" fallacy.
- *Wide* ($m \approx K/2$, $\Delta\theta_{\mathrm{ML}} \sim \pi$). Fixed pointings give at most $\ln(2N) \approx 2.08$ nats; adaptive binary search gives $\ln K \approx 2.77$ nats. Adaptive advantage $= \ln(K/(2N)) = \ln 2 \approx 0.69$ nats *equals* $\mathbb{E}[IG]$ here—EVPI is **tight** and adaptation realizes the full gap.

The narrow regime maximizes the *computable* EVPI upper bound but fails to saturate it; the wide regime has smaller $\mathbb{E}[IG]$ but saturates it fully. Joint optimization favors the wide design: both $V^\star_{\text{fixed}} = \ln(2N)$ and $V^\star_{\text{adaptive}} = \ln K$ are maximized by $m \approx K/2$.

9.4 Full Bellman DP is tractable

This is a finite-state, finite-action, finite-observation POMDP with short horizon. For the terminal reward $\Phi = \ln K - H(b_{N+1})$ (nonlinear in b), the Sondik PWLC representation does not apply directly, but direct backward induction on the reachable belief tree does: from the uniform prior, binary observations under each of K actions produce a tree of at most $(2K)^N$

leaves. At $K = 16, N = 4$ this is $\sim 10^6$ leaves—trivially tabulable. Backward induction with memoization runs in seconds on a laptop. Symmetry reduction by the dihedral D_K group ($|G| = 32$) further reduces the work by up to an order of magnitude.

9.5 Policy iteration for the joint optimum

With exact inner DP available, the envelope-theorem scheme of Section 8 becomes a concrete algorithm:

1. **Policy step.** For current $\boldsymbol{c}_t$, solve the finite POMDP exactly to obtain $\pi^\star(\boldsymbol{c}_t)$ as a depth-N policy tree: every branch indexed by $y_{1:k-1}$ maps to an action θ_k.
2. **Geometry step.** Φ depends on $\boldsymbol{c}$ analytically through (38) and (39). Compute

$$\nabla_{\boldsymbol{c}} W_1\big|_{\boldsymbol{c}_t} \;=\; \mathbb{E}_{z, y_{1:N} \sim \pi^\star(\boldsymbol{c}_t)}\big[\nabla_{\boldsymbol{c}} \Phi\big(z, \pi^\star(y_{1:N}; \boldsymbol{c}_t), \boldsymbol{c}\big)\big]_{\boldsymbol{c}=\boldsymbol{c}_t} \tag{44}$$

   by Monte Carlo over $(z, y_{1:N})$ trajectories under the frozen $\pi^\star(\boldsymbol{c}_t)$. The envelope theorem means no policy Jacobian $d\pi^\star/d\boldsymbol{c}$ is needed.
3. **Gradient step.** Update $\boldsymbol{c}_{t+1} = \mathrm{Proj}_{\mathcal{C}}(\boldsymbol{c}_t + \eta \nabla_{\boldsymbol{c}} W_1)$ and repeat.

Because step 1 is *exact*, the gradient direction (44) is exact—only Monte Carlo variance over trajectories shrinks as $|\text{samples}|^{-1/2}$. Convergence is governed by the smoothness of $W_1(\cdot)$ in $\boldsymbol{c}$: continuous except on codimension-one policy-switch boundaries, through which a subgradient substitutes for the gradient.

9.6 Quantitative demonstration of the necessary-but-not-sufficient property

In the companion numerics document (with code in the same directory), we sweep $\boldsymbol{c}$ over the Gaussian beamwidth $w \in (0, \pi]$ and compute exact V_{oracle}, V_{fixed}, V_{adaptive}, $\mathbb{E}[IG]$ at each w. The in-family optimum for V_{adaptive} is $w^\star = 0.60$ rad with $V^\star_{\text{adaptive}} = 0.87$ nats; the EVPI-maximizing w is the small-w extreme with $V_{\text{adaptive}} = 0.62$ nats—a $1.4\times$ ratio. Within the top-hat family (integer $m \in \{1, \ldots, K\}$), the in-family optimum is $m^\star = 9$ with $V^\star_{\text{adaptive}} = 1.70$ nats, while EVPI-maximizing m gives $V_{\text{adaptive}} = 0.62$ nats—a $2.8\times$ ratio. The largest-EVPI designs in each family are the *smallest*-V_{adaptive} designs, quantitatively confirming Remark 3 (and the analog of the joint-optimum finding in §8.4).

9.7 Adjacent tractable testbeds

The same template covers several other finite-POMDP problems with meaningful geometry:

- *Sequential material identification by spectroscopy*: finite hypothesis set, excitation wavelengths, quantized absorbance; $\boldsymbol{c}$ = spectrometer linewidth and grid.
- *Adaptive compressed sensing with known sparsity*: discrete support set, dictionary rows as actions, thresholded projections as observations; $\boldsymbol{c}$ = dictionary design.
- *Sequential medical-imaging slicing*: discrete lesion-location hypotheses, scan parameters as actions, quantized intensities as observations; $\boldsymbol{c}$ = field-of-view and gradient design.

10 When the joint DP is intractable: a hierarchy of principled relaxations

The radar case study of Section 9 is engineered for exact Bellman tractability: finite state, finite observation alphabet, finite action grid, short horizon. Many sensor-design problems of practical interest violate at least one of these. When exact Bellman is beyond reach, the framework of Sections 7–8 remains the correct organizing picture; only the inner optimization needs approximation. This section catalogs principled relaxations, ordered by structure preserved.

10.1 Sources of intractability

Three structural barriers recur:

1. **Belief-space size.** Continuous $\boldsymbol{x}$ yields an infinite-dimensional belief; one must discretize (grid), project onto a parametric family (e.g., Gaussian), or carry a particle representation.
2. **Action-space size.** Continuous $\boldsymbol{s}$ (beamwidth, amplitude taper, phase profile) turns the inner $\max_{\boldsymbol{s}_k}$ into a nonconvex continuous optimization.
3. **Per-step cost.** Physics-based forward models (FDFD, FDTD, RCWA) make each belief-tree node expensive.

Any one collapses the exact DP; in realistic photonic / metasurface settings all three bite.

10.2 Relaxation path 1: neural adaptive policies (deep RL)

Parametrize the policy as a neural network π_θ and train θ jointly with $\boldsymbol{c}$ by policy-gradient / REINFORCE / pathwise-gradient methods. The joint objective is

$$\max_{\boldsymbol{c},\,\theta}\ \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}\big[\,\Phi(\boldsymbol{x},\,\boldsymbol{s}_{1:N}^{\pi_\theta},\,\boldsymbol{c})\,\big]. \tag{45}$$

What this preserves. Multi-step reasoning via NN memory of the history; arbitrary action geometry; pathwise $\partial\Phi/\partial\boldsymbol{c}$ if the forward model is differentiable. Modern end-to-end optical learning (Sitzmann et al., 2018; Chang & Wetzstein, 2019) fits this template with π_θ collapsed to a one-step reconstruction network.

What this gives up. Optimality certificates: π_θ is an approximator, not a Bellman solver. The envelope theorem (37) applies cleanly only at an interior optimum of θ, rarely attained in practice.

Failure mode: hardware-light fixed points. A familiar pathology of end-to-end optical learning: π_θ can absorb the training signal and compensate for a poor $\boldsymbol{c}$ by investing in policy complexity, leaving $\boldsymbol{c}$ near initialization. The mechanism is gradient asymmetry—$\partial\Phi/\partial\theta$ propagates through cheap high-rank NN backprop, $\partial\Phi/\partial\boldsymbol{c}$ through an expensive low-rank forward-model adjoint. In regimes where geometry has first-order impact on V_{adaptive} (Section 9 quantifies this as $\sim 2.8\times$ in the radar case), this failure mode silently returns a strictly worse answer than the bi-level paths below. Standard countermeasures: (i) capacity control on π_θ; (ii) two-timescale updates with $\boldsymbol{c}$ faster than θ; (iii) warm-start θ from a prescribed information-theoretic rule (2b)/(2c) below; (iv) regularizers penalizing "easy" $\boldsymbol{c}$. In our experience (iii) is most reliable.

10.3 Relaxation path 2: bi-level optimization with a prescribed inner policy

A cheaper, more interpretable route prescribes π in closed form as an information-theoretic rule and keeps the outer envelope-theorem loop of Section 8:

$$\max_{\boldsymbol{c}}\ \mathbb{E}\big[\,\Phi(\boldsymbol{x},\,\boldsymbol{s}_{1:N}[b_\cdot;\boldsymbol{c}],\,\boldsymbol{c})\,\big] \qquad \text{subject to } \boldsymbol{s}_k = \mathsf{R}_k(b_k;\boldsymbol{c}). \tag{46}$$

The outer geometry gradient flows through the implicit $\boldsymbol{c}$-dependence of the inner argmax via the implicit function theorem (IFT)—exact even when the inner rule is not Bellman-optimal. The rule R_k determines how much Bellman structure survives.

(2a) Oracle certainty-equivalent control

At step k, form $\hat{\boldsymbol{x}}_k = \operatorname{argmax}_{\boldsymbol{x}} b_k$ (MAP) or $\hat{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x} \mid b_k]$, then play the first step of the oracle schedule at $\hat{\boldsymbol{x}}_k$:

$$\boldsymbol{s}_k^{\mathrm{CE}} = \big(\boldsymbol{s}^\star(\hat{\boldsymbol{x}}_k; \boldsymbol{c})\big)_1, \qquad \boldsymbol{s}^\star(\boldsymbol{x}; \boldsymbol{c}) = \operatorname{argmax}_{\boldsymbol{s}_{1:N-k+1}} \Phi(\boldsymbol{x},\, \boldsymbol{s}_{1:N-k+1},\, \boldsymbol{c}). \tag{47}$$

Receding-horizon, non-myopic, but assumes $\hat{\boldsymbol{x}}_k$ is correct. Characteristic failure: narrow-beam geometries where $\boldsymbol{s}^\star(\boldsymbol{x})$ collapses to a single pointing at $\boldsymbol{x}$ induce exploration deficits (point-and-verify with a wrong guess). Computationally, one oracle solve per step—if V_{oracle} is the bottleneck, this relaxation does not escape it.

(2b) Myopic one-step information-maximization

Choose each $\boldsymbol{s}_k$ to maximize the expected posterior entropy reduction from the next observation alone:

$$\boldsymbol{s}_k^{\mathrm{myopic}} \;=\; \operatorname{argmax}_{\boldsymbol{s}_k}\; I(\boldsymbol{x};\, y_k \mid b_k, \boldsymbol{s}_k, \boldsymbol{c}). \tag{48}$$

No look-ahead beyond one step. Greedy information gain is $(1 - 1/e)$-competitive against the clairvoyant DP optimum for submodular objectives (Krause & Guestrin, 2008; Golovin & Krause, 2011), the default baseline in sequential design.

(2c) Gaussian-posterior / Fisher-trace surrogate with recursive inference

In the Gaussian-posterior / BvM regime of Section 2.5, $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) \approx \frac{1}{2} \ln\det \boldsymbol{J}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) + \mathrm{const}$, where $\boldsymbol{J}$ is the accumulated Fisher information. A stack of standard approximations on top of (2b) yields the cheapest member of the family:

$$\boldsymbol{s}_k^{\mathrm{BFIM}} \;=\; \operatorname{argmax}_{\boldsymbol{s}_k}\; \operatorname{tr} \boldsymbol{J}\big(\bar{\boldsymbol{x}}_k,\, \boldsymbol{s}_k,\, \boldsymbol{c}\big), \qquad \bar{\boldsymbol{x}}_k = \mathbb{E}[\boldsymbol{x} \mid b_k], \tag{49}$$

coupled with an extended Kalman filter for the belief update. The three stacked simplifications relative to (2b) are:

(i) tr in place of $\ln\det$ (A-optimal, cheaper; loses scale-invariance).
(ii) Evaluate $\boldsymbol{J}$ at $\bar{\boldsymbol{x}}_k$ instead of $\mathbb{E}_{\boldsymbol{x}|b_k}[\boldsymbol{J}(\boldsymbol{x}, \cdot)]$ (Laplace-type approximation).
(iii) EKF for b_k (Gaussian posterior, linearized updates).

Each individually is standard. Stacked, they reduce the inner step to an analytic max-of-trace at a single $\boldsymbol{x}$-point, with a closed-form IFT adjoint for $\boldsymbol{c}$-gradients.

10.4 The hierarchy

Two observations. First, *every row* is a bi-level problem in $\boldsymbol{c}$: the inner step is a function of $(b_k, \boldsymbol{c})$, and the geometry gradient flows through by IFT. The joint $(\boldsymbol{c}, \pi)$ framework of Section 8 is structurally identical across rows; only inner fidelity changes. Second, path 1 (deep RL) and paths 2a–2c are not mutually exclusive: a deep-RL policy can be warm-started from a prescribed rule, combining the sample efficiency of (2b)–(2c) with the flexibility of (1).

10.5 Transition: the photonic sensor problem as an instance of (2c)

A companion body of work develops a worked instance of path (2c) applied to a 2D photonic sensor: a cross-waveguide device whose permittivity distribution is the geometry variable $\boldsymbol{c}$, whose sensor parameters are input-port phases, and whose observations are port powers. Each of the three bulleted simplifications (i)–(iii) above is justified by the operating regime:

- The state $\boldsymbol{x}$ is a small refractive-index perturbation ($\Delta n \in [10^{-5}, 10^{-4}]$), so the posterior remains unimodal and near-Gaussian—justifying (iii).

Table 1: Tractability hierarchy for adaptive sensor design. Every row is a bi-level problem in $\boldsymbol{c}$; only the fidelity of the inner optimization changes. The exact DP is the gold standard the radar case study of Section 9 exhibits.

	inner step per $\boldsymbol{s}_k$	look-ahead	posterior handling	guarantee
Exact Bellman DP	full belief backup	N-step	exact	joint-optimal
Deep RL (§10.2)	one NN forward pass	$\leq N$, implicit	NN-internal	approximate, black-box
Oracle CE (2a)	oracle at $\hat{\boldsymbol{x}}_k$	$(N-k+1)$-step CE	point estimate	none; under-explores
Myopic info-gain (2b)	$\operatorname{argmax} I(\boldsymbol{x}; y_k \mid b_k)$	1-step	full posterior	$(1-1/e)$ via submodularity
Fisher-trace (2c)	$\operatorname{argmax} \operatorname{tr} \boldsymbol{J}(\bar{\boldsymbol{x}}_k)$	1-step	Gaussian / EKF	asymptotically exact under BvM

- The Fisher information $\boldsymbol{J}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ varies slowly in $\boldsymbol{x}$ over the posterior support—justifying (ii).
- $\operatorname{tr} \boldsymbol{J}$ admits an analytically differentiable inner argmax—justifying (i).

The photonic problem is not an alternative *framework*; it is an instantiation of the joint-$(\boldsymbol{c}, \pi)$ framework of this paper at row (2c), with each approximation justified by the physics of its operating regime. The radar case study of Section 9 and the photonic case study are two points on the same methodological spectrum: exact DP where tractable, (2c) where not. The message of both is the same: *joint* optimization of $(\boldsymbol{c}, \pi)$ is the correct object; the inner solver depends on what is affordable.

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A Identities for the pointwise information gain Φ

Three equivalent forms of $\Phi(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ in Definition 1:

Entropy-reduction form (primary).

$$\Phi \;=\; H(p_0) - \mathbb{E}_{\boldsymbol{y}\sim p(\cdot|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c})}[H(p(\cdot \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c}))]. \tag{50}$$

Expected-KL form.

$$\Phi \;=\; \mathbb{E}_{\boldsymbol{y}\sim p(\cdot|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c})}\big[\mathrm{KL}\big(p(\cdot \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c}) \,\big\|\, p_0\big)\big]. \tag{51}$$

Log-likelihood-ratio form.

$$\Phi \;=\; \mathbb{E}_{\boldsymbol{y}\sim p(\cdot|\boldsymbol{x},\boldsymbol{s},\boldsymbol{c})}\left[\int p(\boldsymbol{x}' \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c}) \ln \frac{p(\boldsymbol{x}' \mid \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{c})}{p_0(\boldsymbol{x}')} d\boldsymbol{x}'\right]. \tag{52}$$

Equivalence follows from $\mathrm{KL}(p \,\|\, p_0) = \int p \ln(p/p_0) = -H(p) - \int p \ln p_0$ combined with $\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}} \int p(\boldsymbol{x}' \mid \boldsymbol{y}) \ln p_0(\boldsymbol{x}') d\boldsymbol{x}' = \mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}} \mathbb{E}_{\boldsymbol{x}'|\boldsymbol{y}}[\ln p_0(\boldsymbol{x}')]$; expanding and simplifying gives the stated identities.

The prior-average identity $\mathbb{E}_{\boldsymbol{x}}[\Phi] = I(\boldsymbol{x}; \boldsymbol{y})$ (Theorem 1) is fundamental to the framework. It means: what the oracle / ignorance-gap / joint-DP machinery does pointwise, the standard mutual-information framework does in aggregate—with the pointwise version strictly richer because it supports oracle conditioning on $\boldsymbol{x}$.

B Derivation of Bellman's equation

We derive equations (31)–(32) from first principles. The argument is general to any finite-horizon sequential decision problem.

B.1 Problem setup

Recall the ingredients:

- An unknown state $\boldsymbol{x} \sim p_0$.
- Horizon N with steps $k = 1, 2, \ldots, N$.
- At step k, an action $\boldsymbol{s}_k$ is chosen; then $\boldsymbol{y}_k \sim p(\cdot \mid \boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})$ is observed.
- An adaptive policy $\pi = (\pi_1, \ldots, \pi_N)$ with $\pi_k : \boldsymbol{y}_{1:k-1} \mapsto \boldsymbol{s}_k$.
- A terminal reward $R(b_{N+1}) = H(p_0) - H(b_{N+1})$ accrued at the end.

Under policy π and initial prior p_0, the expected reward is

$$V^{\pi}(\boldsymbol{c}) \;=\; \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}\big[R(b_{N+1})\big]. \tag{53}$$

By construction, $V^{\pi}(\boldsymbol{c}) = \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}_{1:N}}[\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}), \boldsymbol{c})]$: terminal-entropy value-to-go equals the expected pointwise information gain.

The goal:

$$V_{\text{adaptive}}(\boldsymbol{c}) \;=\; \max_{\pi} V^{\pi}(\boldsymbol{c}). \tag{54}$$

B.2 Sufficient statistic: the belief

All information in $\boldsymbol{y}_{1:k-1}$ relevant to future decisions is carried by the posterior

$$b_k(\boldsymbol{x}) \;=\; p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1}, \boldsymbol{s}_{1:k-1}, \boldsymbol{c}). \tag{55}$$

Two histories yielding the same b_k are indistinguishable going forward, so without loss we search over policies $\pi_k : b_k \mapsto \boldsymbol{s}_k$.

Bayesian updating: given $(b_k, \boldsymbol{s}_k, \boldsymbol{y}_k)$,

$$b_{k+1}(\boldsymbol{x}) \;=\; \frac{p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\, b_k(\boldsymbol{x})}{\int p(\boldsymbol{y}_k \mid \boldsymbol{x}', \boldsymbol{s}_k, \boldsymbol{c})\, b_k(\boldsymbol{x}')\, d\boldsymbol{x}'}. \tag{56}$$

B.3 Value-to-go

Define

$$W_k(b_k; \boldsymbol{c}) \;=\; \max_{\pi_k,\ldots,\pi_N} \mathbb{E}\big[R(b_{N+1}) \mid b_k, \pi_{k:N}\big]. \tag{57}$$

Boundary cases:

- $k = 1$: $W_1(p_0; \boldsymbol{c}) = V_{\text{adaptive}}(\boldsymbol{c})$.
- $k = N + 1$: no decisions remain; $W_{N+1}(b_{N+1}; \boldsymbol{c}) = R(b_{N+1}) = H(p_0) - H(b_{N+1})$.

B.4 Peeling off one step

Split the outer max:

$$W_k(b_k; \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k} \max_{\pi_{k+1},\ldots,\pi_N} \mathbb{E}[R \mid b_k, \boldsymbol{s}_k, \pi_{k+1:N}]. \tag{58}$$

Apply the tower property, conditioning on $\boldsymbol{y}_k$ (which determines b_{k+1} given $(b_k, \boldsymbol{s}_k)$):

$$\mathbb{E}[R \mid b_k, \boldsymbol{s}_k, \pi_{k+1:N}] \;=\; \mathbb{E}_{\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k}\big[\,\mathbb{E}[R \mid b_{k+1}, \pi_{k+1:N}]\,\big]. \tag{59}$$

Pushing $\max_{\pi_{k+1:N}}$ inside $\mathbb{E}_{\boldsymbol{y}_k}$ (justified in the next subsection):

$$W_k(b_k; \boldsymbol{c}) \;=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k}\big[\max_{\pi_{k+1:N}} \mathbb{E}[R \mid b_{k+1}, \pi_{k+1:N}]\big] \;=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k}\big[W_{k+1}(b_{k+1}; \boldsymbol{c})\big]. \tag{60}$$

The boxed equation (31) is (60).

B.5 Why the push-in of the max is legal

In general $\mathbb{E}[\max_i X_i] \geq \max_i \mathbb{E}[X_i]$—strict inequality. So why is the push-in legal here? Because $\pi_{k+1:N}$ may *depend on* $\boldsymbol{y}_k$ (through b_{k+1}): a policy that observes $\boldsymbol{y}_k$ before committing the continuation is strictly more powerful than one that commits in advance, and the problem allows the former. Formally: maximizing pointwise inside the expectation gives the same value as maximizing over all measurable continuation plans outside. This is the *interchangeability lemma* for adaptive optimization—the rigorous form of Bellman's *principle of optimality.*

B.6 The Bellman system

$$
\begin{aligned}
W_{N+1}(b_{N+1};\boldsymbol{c}) &= H(p_0) - H(b_{N+1}), && (61)\\
W_k(b_k;\boldsymbol{c}) &= \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k,\boldsymbol{c}}\big[W_{k+1}(b_{k+1};\boldsymbol{c})\big], \quad k = N,\dots,1, && (62)\\
\pi_k^\star(b_k;\boldsymbol{c}) &= \operatorname{argmax}_{\boldsymbol{s}_k} \ \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k,\boldsymbol{c}}\big[W_{k+1}(b_{k+1};\boldsymbol{c})\big], && (63)\\
V_{\text{adaptive}}(\boldsymbol{c}) &= W_1(p_0;\boldsymbol{c}). && (64)
\end{aligned}
$$

B.7 Summary

Three ingredients yield the Bellman equation:

(a) Sufficient statistic reduction: replace $\boldsymbol{y}_{1:k-1}$ with b_k.

(b) Peeling: $\max_{\pi_{k:N}} = \max_{\boldsymbol{s}_k} \max_{\pi_{k+1:N}}$.

(c) Tower property with interchangeability: push $\max_{\pi_{k+1:N}}$ inside $\mathbb{E}_{\boldsymbol{y}_k}$.

The recursion rests on two facts: the Markov property of beliefs under Bayesian updating, and the principle of optimality (the optimal continuation can depend on intervening observations without loss).